

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations
 December 2024 - January 2025

22MCA101 – DATA STRUCTURES WITH ALGORITHMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from each **Unit**.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Explain various data types in C language with syntax and examples.	10	L*2	1	1
	b)	Illustrate a program to store 10 different numbers in a Single Dimensional Array (SDA).				
	i)	To display all the number in reverse order				
	ii)	To display the sum and average of all the numbers	10	L2	1	2
2.	a)	Explain stack with an example. Implement push and pop operations for stack using arrays.	10	L2	1	1
	b)	Illustrate a C function to convert infix to a postfix expression. Convert the given expression $((A + B) - C * (D / E)) + F$ into its equivalent postfix.	10	L2	1	2
		Unit – II				
3.	a)	Illustrate recursive definition and process for generating Fibonacci series.	10	L2	2	2
	b)	Develop a C program to implement insertion, deletion and display operations on Linear queue.	10	L3	2	2
4.	a)	Compare linear queue vs circular queue, and implement a circular queue using an array.	10	L2	2	1
	b)	Develop a recursive function to solve the Towers of Hanoi problem for n disks and illustrate it for n=4.	10	L3	2	2
		Unit – III				
5.	a)	Explain the following with suitable functions:				
	i)	Inserting a node at the beginning of a Doubly linked list	10	L2	3	1
	ii)	Deleting a node at the end of the Doubly linked list				
	b)	Develop a C program for addition of long integers using doubly linked lists.	10	L3	3	2
6.	a)	Explain the key differences between singly linked lists, doubly linked lists, and circular linked lists with the help of suitable examples. Discuss the advantages and applications of each type.	10	L2	3	1
	b)	Develop C functions for the following:				
	i)	Find the maximum element in the singly linked list	10	L3	3	2
	ii)	Mean of elements in singly linked list				
		Unit – IV				
7.	a)	Explain Algorithm design and analysis process with a neat diagram and an example.	10	L2	4	1
	b)	Derive asymptotic notations (Big-O and Big- Omega) for the below functions.				
	i)	$f(n) = 2n^3 + 4n + 5$				
	ii)	$f(n) = 3n^2 + 2n + 4 \leq 4n^2$	10	L2	4	2
8.	a)	Explain the concept of Threaded Binary Trees in detail. Discuss the types, of threaded binary trees, their structure, and advantages over regular binary trees.	10	L2	4	1

P.T.O.

- b) Illustrate the different tree traversal techniques used in binary trees. Also provide the detailed explanation of the in-order, pre-order, and post-order traversals with examples.

10 L2 4 1

Unit – V

9. a) Explain all pair shortest path for the diagraph with the following weight matrix:

$$\begin{bmatrix} 0 & 2 & \infty & 1 & 8 \\ 6 & 0 & 3 & 2 & \infty \\ \infty & \infty & 0 & 4 & \infty \\ \infty & \infty & 2 & 0 & 3 \\ 3 & \infty & \infty & \infty & 0 \end{bmatrix}$$

10 L2 4 2

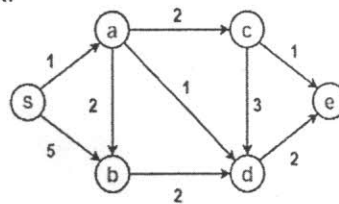
- b) Apply quick sort algorithm to sort the list A, L, G, O, R, I, T, H, M in alphabetical order. Draw the tree of recursive calls made.

10 L3 5 2

10. a) Illustrate the problem-solving techniques through the development of an algorithm for identifying the position of a target element within a sorted array of integers.

10 L2 5 1

- b) Apply single source shortest path algorithm for given graph by considering 'S' as the source vertex.



10 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
